A6 · Answer Key and Demo Script

CS425 · Who Carries Your Packets? · instructor led

INSTRUCTOR COPY

Every transcript below is real output captured on `onyx.boisestate.edu` on October 4, 2026. Hops, times and the RIPEstat counts will drift; the shapes will not. About 40 minutes: 8, 10, 12, 6, then 4 for the exit question.

Say this at the start

"The Internet is not one network, it is more than 70,000 of them, and Boise State is one. Today we find out which one, who it pays to reach everybody else, and what the rest of the world sees when it looks back at us. Everyone `ssh onyx`." Then hand out the sheet.

ROUND 1 · WHO OWNS ONYX'S ADDRESS?

dig +short TXT 11.227.178.132.origin.asn.cymru.com then AS46662.asn.cymru.com

```
"46662 | 132.178.0.0/16 | US | arin | 1988-12-05"
"46662 | US | arin | 2008-11-07 | BSU-AS - Boise State University, US"
```

SAY "The address goes in backwards, like a reverse DNS lookup, because DNS names get more specific to the left."

Prefix. AS 46662, BSU-AS, announces `132.178.0.0/16`: $2^{16} = 65{,}536$ addresses. Shorter than the /25, and it covers Onyx's /25 and every other campus subnet.

Q1. Aggregation: one route for the whole campus keeps every BGP table on the Internet smaller, and the outside world does not need to know how campus carves up its block. Only routers **inside** AS 46662 know the /25, and they learn it from the intra-AS protocol (OSPF or similar, 5.3). BGP carries only the /16 (5.4).

ROUND 2 · MAP THE TRACE ONTO THE ASES

mtr -n -z -T -P 443 -r -c 3 mirrors.xmission.com

```
  1. AS46662  132.178.227.1        0.0%     3    0.7   0.6   0.6   0.7   0.0
  2. AS???    ???                100.0     3    0.0   0.0   0.0   0.0   0.0
  3. AS46662  132.178.1.147        0.0%     3    0.6   0.7   0.6   0.9   0.1
  4. AS46662  132.178.1.98         0.0%     3    1.3   1.4   1.3   1.5   0.1
  5. AS6461   209.249.232.13       0.0%     3    1.3   1.7   1.2   2.6   0.8
  6-9. AS???  ???                100.0
 10. AS3356   4.69.220.109         0.0%     3   27.8  27.8  27.2  28.4   0.6
 11. AS3356   4.35.168.186         0.0%     3   51.7  52.2  51.7  52.5   0.4
 12. AS6315   166.70.1.2           0.0%     3   52.2  52.0  51.8  52.2   0.2
 13. AS6315   198.60.22.13         0.0%     3   51.8  51.9  51.8  52.0   0.1
```

Count. 4 ASes: 46662 Boise State, 6461 Zayo, 3356 Lumen (still registered as LEVEL3), 6315 XMission. AS path: `46662 6461 3356 6315`. BGP writes the AS-PATH newest first with the origin last, so the route Boise State's routers hold for XMission reads `6461 3356 6315`.

Q2. Crossings: hop 4 to 5 (Boise State to Zayo), somewhere in 5 to 10 (Zayo to Lumen, hidden by the silent hops), 11 to 12 (Lumen to XMission). Hops 1 to 4 are inside Boise State, 10 to 11 inside Lumen, 12 to 13 inside XMission; the silent hops 6 to 9 are probably inside Zayo. Twelve routers, four ASes: BGP only ever sees the four.

ROUND 3 · BOISE STATE FROM THE OUTSIDE

RIPEstat asn-neighbours for AS46662, then the looking glass for 132.178.0.0/16

```
{"asn":46435,"type":"left","power":33,"v4_peers":43,"v6_peers":0}
{"asn":6461,"type":"left","power":223,"v4_peers":292,"v6_peers":0}
"46435 | US | arin | 2008-09-17 | IRON - IRON, US"

     10 24482 6461 46662
      4 6461 46662
      4 14840 3356 46435 46662
     ...
     40 46435
    258 6461
```

SAY "Read a path right to left: that is the order the route was passed along, starting from us."

Neighbors. Two, both providers: AS 6461 Zayo and AS 46435 IRON (Idaho Regional Optical Network). Of 298 paths, 258 come in through Zayo and 40 through IRON.

Q3. Two providers is **multihoming**: if one link or provider fails, BGP withdraws those routes and traffic shifts to the other. No `right` neighbors means nobody reaches the Internet through Boise State and it announces only its own /16: it is a **stub** AS, not a transit AS, and its providers are paid to carry its traffic. A route whose AS-PATH already holds 6461 has been through Zayo once, so Zayo **rejects** it: that is AS-PATH loop detection (5.4).

FAST STUDENT Why do most of the world's routers pick Zayo? Route selection is local preference first, then path length. IRON is a regional network that buys its own transit (its paths run through Lumen, `3356 46435 46662`), so the route through IRON is usually one AS longer than the route through Zayo. That is the likely story, not something the data proves.

ROUND 4 · NEAR IS NOT FAST

mtr -n -z -T -P 443 -r -c 3 8.8.8.8

```
  1. AS46662  132.178.227.1        0.0%     3    0.7   0.6   0.6   0.7   0.0
  3-4. AS46662  132.178.1.147, 132.178.1.98
  5. AS6461   209.249.232.13       0.0%     3    1.2   1.3   1.2   1.4   0.1
  6-8. AS???  ???                 100.0
  9. AS15169  142.250.169.166      0.0%     3   11.3  11.4  11.3  11.5   0.1
10-11. AS15169  (two or three addresses per hop)
12. AS15169  8.8.8.8              0.0%     3   11.5  15.5  11.5  18.6   3.6
```

Answers. Google, about 11 ms, versus about 52 ms to Salt Lake City. AS path `46662 6461 15169`: 3 ASes, so Zayo hands traffic straight to Google.

Q4. 11 ms round trip is at most about 1,100 km each way, and less in practice. Because 8.8.8.8 is **anycast**, you reach whichever copy BGP gives you the best route to: Zayo hears 8.8.8.8 from Google in many places and picks one by its own policy. On the XMission path, 27 ms at the first Lumen router means the packet went a long way before Zayo handed it to Lumen, probably to some city where the two connect, then back to Salt Lake. BGP picks paths by policy and AS hops, not by kilometers. The `mtr` lines with more than one address at hops 10 and 11 are Google balancing load across parallel routers.

EXIT QUESTION

Local preference. Boise State gives routes learned from IRON a higher local preference, and every router in the AS prefers them. Local preference is the **first** rule in route selection, so it is decided before AS-PATH length is ever compared. BGP is policy first: who you pay, and how much, beats how short the path is.

GRADING

Pass/fail, 20 points. Pass when every box has something in it. Wrong predictions are fine; that is what they are for.